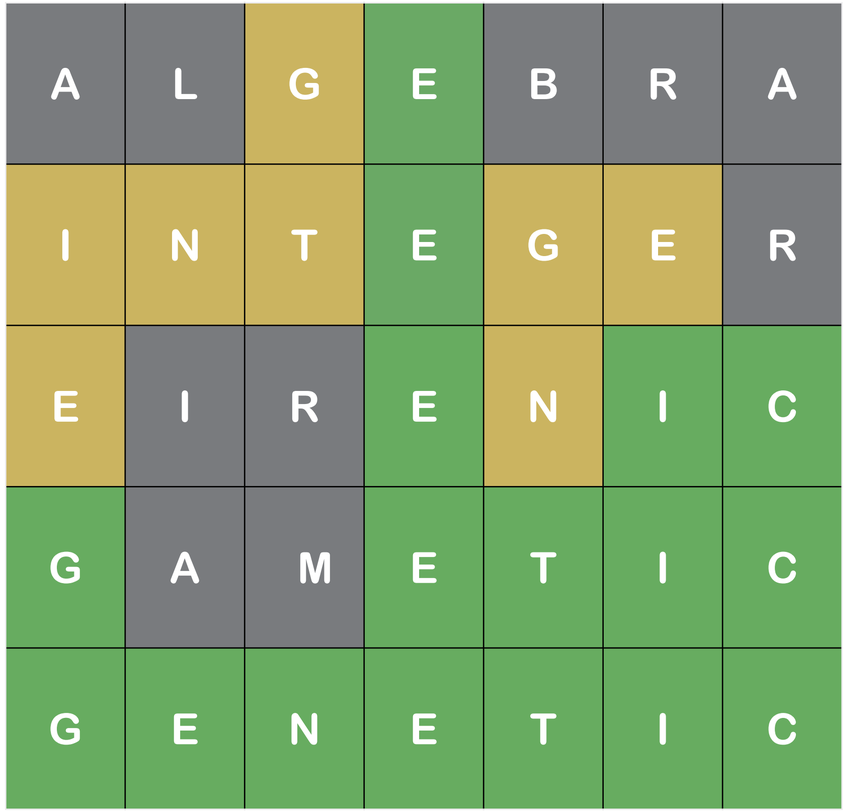

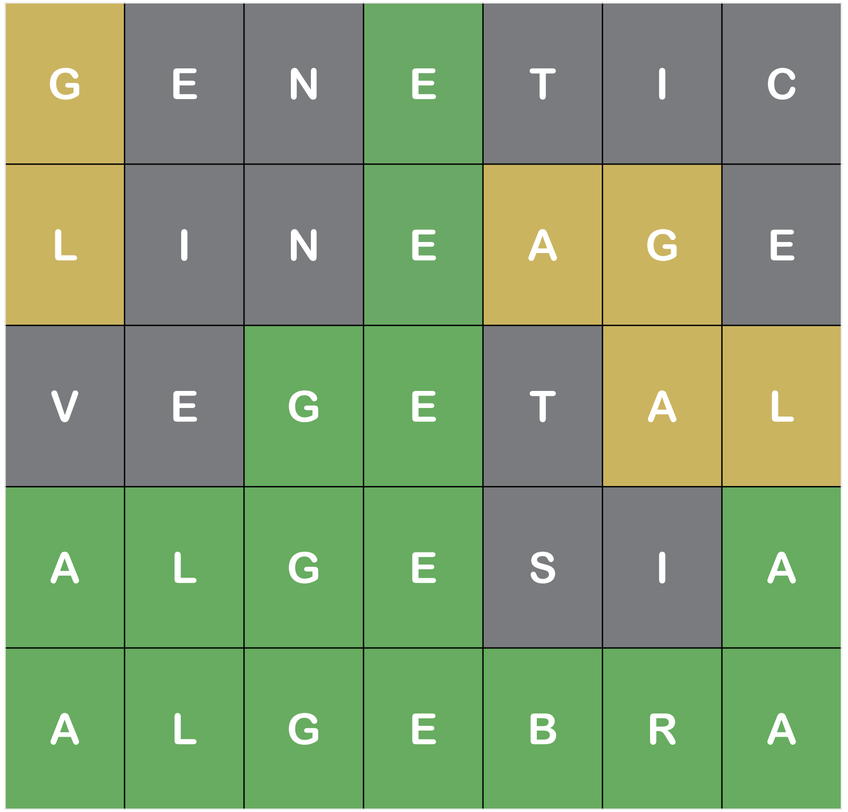

the algebraic structure of genetic inheritance

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In Mendelian **genetics**, the mechanisms of inheritance are represented by **algebraic symbols**. For 2 alleles there are 3 **zygotic genotypes**.

	AA	Aa	aa
AA	AA	$\frac{1}{2}(AA + Aa)$	Aa
Aa	$\frac{1}{2}(AA + Aa)$	$\frac{1}{4}AA + \frac{1}{2}Aa + \frac{1}{4}aa$	$\frac{1}{2}(Aa + aa)$
aa	Aa	$\frac{1}{2}(Aa + aa)$	aa

Algebra, geometry, topology, analysis, and combinatorics, are branches of mathematics.

What is an **algebra**?

An **algebra** (over a field) is a **vector space** with a **bilinear product**

A **vector space** is a nonempty set V of objects, called *vectors*, on which are defined two operations, called *addition* and *multiplication by scalars* (real numbers), subject to the ten axioms (or rules) listed below.[1] The axioms must hold for all vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ in V and for all scalars c and d.

1. The sum of $\mathbf{u}$ and $\mathbf{v}$, denoted by $\mathbf{u}+\mathbf{v}$, is in V.
2. $\mathbf{u}+\mathbf{v}=\mathbf{v}+\mathbf{u}$.
3. $(\mathbf{u}+\mathbf{v})+\mathbf{w}=\mathbf{u}+(\mathbf{v}+\mathbf{w})$.
4. There is a **zero** vector $\mathbf{0}$ in V such that $\mathbf{u}+\mathbf{0}=\mathbf{u}$.
5. For each $\mathbf{u}$ in V, there is a vector $-\mathbf{u}$ in V such that $\mathbf{u}+(-\mathbf{u})=\mathbf{0}$.
6. The scalar multiple of $\mathbf{u}$ by c, denoted by $c\mathbf{u}$, is in V.
7. $c(\mathbf{u}+\mathbf{v})=c\mathbf{u}+c\mathbf{v}$.
8. $(c+d)\mathbf{u}=c\mathbf{u}+d\mathbf{u}$.
9. $c(d\mathbf{u})=(cd)\mathbf{u}$.
10. $1\mathbf{u}=\mathbf{u}$.

$$\mathbf{uv} \in V$$

$$(c\mathbf{u}+d\mathbf{v}) = c\mathbf{uw}+d\mathbf{vw}$$

$$\mathbf{w}(c\mathbf{u}+d\mathbf{v}) = c\mathbf{wu}+d\mathbf{wv}$$

In algebras relevant to genetics

multiplication is **commutative**

$$\mathbf{uv} = \mathbf{vu}$$

but NOT associative

$$(\mathbf{uv})\mathbf{w} \neq \mathbf{u}(\mathbf{vw})$$

Genetic algebras are **not** associative

Population U mates with population V and the resulting population mates with W

$$(\mathbf{uv})\mathbf{w} \neq \mathbf{u}(\mathbf{vw})$$

Population V mates with population W and the resulting population mates with U

---

multiplication table for the **gametic algebra** of Mendelian inheritance

	A	a
A	A	$\frac{1}{2}(A+a)$
a	$\frac{1}{2}(a+A)$	a

$$\begin{aligned}(\mathsf{A}\cdot\mathsf{A})\cdot\mathsf{a} &= \mathsf{A}\cdot\mathsf{a}\\ &= \tfrac{1}{2}\mathsf{A}+\tfrac{1}{2}\mathsf{a}\end{aligned}$$

$$\begin{aligned}\mathsf{A}\cdot(\mathsf{A}\cdot\mathsf{a}) &= \mathsf{A}\cdot(\tfrac{1}{2}\mathsf{A}+\tfrac{1}{2}\mathsf{a})\\ &= \tfrac{1}{2}\mathsf{A}\cdot\mathsf{A}+\tfrac{1}{2}\mathsf{A}\cdot\mathsf{a}\\ &= \tfrac{1}{2}\mathsf{A}+\tfrac{1}{2}(\tfrac{1}{2}\mathsf{A}+\tfrac{1}{2}\mathsf{a})\\ &= \tfrac{1}{2}\mathsf{A}+\tfrac{1}{4}\mathsf{A}+\tfrac{1}{4}\mathsf{a}\\ &= \tfrac{3}{4}\mathsf{A}+\tfrac{1}{4}\mathsf{a}\end{aligned}$$

Example: self fertilization

Zygotic distribution: $x = \alpha\mathsf{AA} + \beta\mathsf{Aa} + \gamma\mathsf{aa} \quad \text{with} \quad \alpha + \beta + \gamma = 1$

$$
\begin{aligned}
x^2 \qquad F_1 &= \alpha(AA \cdot AA) + \beta(Aa \cdot Aa) + \gamma(aa \cdot aa) \\
&= \alpha AA + \beta\left(\frac{1}{4}AA + \frac{1}{2}Aa + \frac{1}{4}aa\right) + \gamma aa \\
&= \left(\alpha + \frac{1}{4}\beta\right)AA + \frac{1}{2}\beta Aa + \left(\frac{1}{4}\beta + \gamma\right)aa.
\end{aligned}
$$

$$
(x^2)^2 \qquad F_2 = \left(\alpha + \frac{3}{8}\beta\right)AA + \frac{1}{4}\beta Aa + \left(\frac{3}{8}\beta + \gamma\right)aa.
$$

$$F_n = \underbrace{(\cdots((x^2)^2)\cdots)^2}_{x \text{ squared } n \text{ times}}$$

$$= \left(\alpha + \frac{1}{2}\beta - \frac{1}{2^{n+1}}\beta\right) AA + \frac{1}{2^n}\beta Aa + \left(\frac{1}{2}\beta + \gamma - \frac{1}{2^{n+1}}\beta\right) aa.$$

Limiting distribution: $\lim_{n\to\infty} F_n = (\alpha + \frac{1}{2}\beta)\mathsf{AA} + (\gamma + \frac{1}{2}\beta)\mathsf{aa}$

Repeated self fertilization eliminates heterozygotes
(50% reduction with each generation)

General form of a genetic algebra

$$\boldsymbol{x} = \sum_{k=1}^{n} x_k \boldsymbol{a}_k = x_1\boldsymbol{a}_1 + x_2\boldsymbol{a}_2 + \cdots + x_n\boldsymbol{a}_n \qquad \sum_{k=1}^{n} x_k = 1$$

general element of vector space

k-th basis element

n = dimension

normalized

$$\boldsymbol{a}_i\boldsymbol{a}_j = \sum_{k=1}^{n} \gamma_{ij}^{k}\boldsymbol{a}_k \qquad 0 \le \gamma_{ij}^{k} \le 1 \qquad \sum_{k=1}^{n} \gamma_{ij}^{k} = 1 \qquad \gamma_{ij}^{k} = \gamma_{ji}^{k}$$

structure constants

	$\boldsymbol{a}_1$	$\boldsymbol{a}_2$	$\boldsymbol{a}_3$
$\boldsymbol{a}_1$	$\gamma_{11}^{1}\boldsymbol{a}_1 + \gamma_{11}^{2}\boldsymbol{a}_2 + \gamma_{11}^{3}\boldsymbol{a}_3$	$\gamma_{12}^{1}\boldsymbol{a}_1 + \gamma_{12}^{2}\boldsymbol{a}_2 + \gamma_{12}^{3}\boldsymbol{a}_3$	$\gamma_{13}^{1}\boldsymbol{a}_1 + \gamma_{13}^{2}\boldsymbol{a}_2 + \gamma_{13}^{3}\boldsymbol{a}_3$
$\boldsymbol{a}_2$	$\star$	$\gamma_{22}^{1}\boldsymbol{a}_1 + \gamma_{22}^{2}\boldsymbol{a}_2 + \gamma_{22}^{3}\boldsymbol{a}_3$	$\gamma_{23}^{1}\boldsymbol{a}_1 + \gamma_{23}^{2}\boldsymbol{a}_2 + \gamma_{23}^{3}\boldsymbol{a}_3$
$\boldsymbol{a}_3$	$\star$	$\star$	$\gamma_{33}^{1}\boldsymbol{a}_1 + \gamma_{33}^{2}\boldsymbol{a}_2 + \gamma_{33}^{3}\boldsymbol{a}_3$

Sequences of interest

repeated back-crossing

$$x, x^2, x^3, x^4, x^5, \dots$$

principal powers

$$x^k = xx^{k-1}$$

repeated random mating

$$x, x^2, (x^2)^2, ((x^2)^2)^2, \dots$$

plenary powers

$$x^{[k]} = x^{[k-1]}x^{[k-1]}$$

descendents of X within population Y

$$x, yx, y(yx), y(y(yx))), \cdots$$

$$y^k x = yy^{k-1}x$$

The **principal rank equation** is the polynomial of lowest degree connecting the **principal powers**

$$f(\boldsymbol{x}) = 0$$

$$f(x) = x^r + \theta_1 x^{r-1} + \theta_2 x^{r-2} + \cdots + \theta_{r-2} x^2 + \theta_{r-1} x$$

$$f(x) = x(x-1)(x-\lambda_1)(x-\lambda_2)\cdots(x-\lambda_{r-2})$$

the principal rank equation for the standard zygotic algebra

$$\boldsymbol{x} = x_{\mathrm{AA}}\mathsf{AA} + x_{\mathrm{Aa}}\mathsf{Aa} + x_{\mathrm{aa}}\mathsf{aa}$$

	AA	Aa	aa
AA	AA	$\frac{1}{2}$AA + $\frac{1}{2}$Aa	Aa
Aa	$\star$	$\frac{1}{4}$AA + $\frac{1}{2}$Aa + $\frac{1}{4}$aa	$\frac{1}{2}$Aa + $\frac{1}{2}$aa
aa	$\star$	$\star$	aa

$$f(\boldsymbol{x}) = \boldsymbol{x}^3 - \boldsymbol{x}^2 = \boldsymbol{x}^2(\boldsymbol{x} - 1) = 0$$

Interpretation: after a single round of random mating the genotype distribution of a population is equilibrated

product of two elements

$\boldsymbol{xy}$

$$(x_{\mathrm{AA}}\mathsf{AA} + x_{\mathrm{Aa}}\mathsf{Aa} + x_{\mathrm{aa}}\mathsf{aa})\,(y_{\mathrm{AA}}\mathsf{AA} + y_{\mathrm{Aa}}\mathsf{Aa} + y_{\mathrm{aa}}\mathsf{aa})$$

	AA	Aa	aa
AA	AA	$\frac{1}{2}$AA + $\frac{1}{2}$Aa	Aa
Aa	$\star$	$\frac{1}{4}$AA + $\frac{1}{2}$Aa + $\frac{1}{4}$aa	$\frac{1}{2}$Aa + $\frac{1}{2}$aa
aa	$\star$	$\star$	aa

	$x_{\mathrm{AA}}\mathsf{AA}$	$x_{\mathrm{Aa}}\mathsf{Aa}$	$x_{\mathrm{aa}}\mathsf{aa}$
$y_{\mathrm{AA}}\mathsf{AA}$	$x_{\mathrm{AA}}y_{\mathrm{AA}}\mathsf{AA}$	$\frac{1}{2}x_{\mathrm{Aa}}y_{\mathrm{AA}}\mathsf{AA} + \frac{1}{2}x_{\mathrm{Aa}}y_{\mathrm{AA}}\mathsf{Aa}$	$x_{\mathrm{aa}}y_{\mathrm{AA}}\mathsf{Aa}$
$y_{\mathrm{Aa}}\mathsf{Aa}$	$\frac{1}{2}x_{\mathrm{AA}}y_{\mathrm{Aa}}\mathsf{AA} + \frac{1}{2}x_{\mathrm{AA}}y_{\mathrm{Aa}}\mathsf{Aa}$	$\frac{1}{4}x_{\mathrm{Aa}}y_{\mathrm{Aa}}\mathsf{AA} + \frac{1}{2}x_{\mathrm{Aa}}y_{\mathrm{Aa}}\mathsf{Aa} + \frac{1}{4}x_{\mathrm{Aa}}y_{\mathrm{Aa}}\mathsf{aa}$	$\frac{1}{2}x_{\mathrm{aa}}y_{\mathrm{Aa}}\mathsf{Aa} + \frac{1}{2}x_{\mathrm{aa}}y_{\mathrm{Aa}}\mathsf{aa}$
$y_{\mathrm{aa}}\mathsf{aa}$	$x_{\mathrm{AA}}y_{\mathrm{aa}}\mathsf{Aa}$	$\frac{1}{2}x_{\mathrm{Aa}}y_{\mathrm{aa}}\mathsf{Aa} + \frac{1}{2}x_{\mathrm{Aa}}y_{\mathrm{aa}}\mathsf{aa}$	$x_{\mathrm{aa}}y_{\mathrm{aa}}\mathsf{aa}$

$$\begin{aligned}
\boldsymbol{xy} = &\left(x_{\mathrm{AA}}y_{\mathrm{AA}} + \frac{x_{\mathrm{Aa}}y_{\mathrm{AA}} + x_{\mathrm{AA}}y_{\mathrm{Aa}}}{2} + \frac{x_{\mathrm{Aa}}y_{\mathrm{Aa}}}{4}\right)\mathsf{AA} \\
&+ \left(x_{\mathrm{aa}}y_{\mathrm{AA}} + x_{\mathrm{AA}}y_{\mathrm{aa}} + \frac{x_{\mathrm{Aa}}(y_{\mathrm{AA}} + y_{\mathrm{aa}}) + x_{\mathrm{Aa}}y_{\mathrm{Aa}} + (x_{\mathrm{AA}} + x_{\mathrm{aa}})y_{\mathrm{Aa}}}{2}\right)\mathsf{Aa} \\
&+ \left(x_{\mathrm{aa}}y_{\mathrm{aa}} + \frac{x_{\mathrm{aa}}y_{\mathrm{Aa}} + x_{\mathrm{Aa}}y_{\mathrm{aa}}}{2} + \frac{x_{\mathrm{Aa}}y_{\mathrm{Aa}}}{4}\right)\mathsf{aa}\,.
\end{aligned}$$

transformation matrix

vectors

$$\boldsymbol{x}\,,\boldsymbol{y}\in\mathcal{A}$$

linear map

$$M_{\boldsymbol{x}}:\boldsymbol{y}\mapsto\boldsymbol{xy}$$

$$\boldsymbol{xy}=M_{\boldsymbol{x}}\boldsymbol{y}$$

$$\boldsymbol{xy}=\begin{pmatrix} x_{\mathrm{AA}}y_{\mathrm{AA}}+\frac{1}{2}(x_{\mathrm{Aa}}y_{\mathrm{AA}}+x_{\mathrm{AA}}y_{\mathrm{Aa}})+\frac{1}{4}Aay_{\mathrm{Aa}} \\ x_{\mathrm{aa}}y_{\mathrm{AA}}+x_{\mathrm{AA}}y_{\mathrm{aa}}+\frac{1}{2}[x_{\mathrm{Aa}}(y_{\mathrm{AA}}+y_{\mathrm{aa}})+x_{\mathrm{Aa}}y_{\mathrm{Aa}}+(x_{\mathrm{AA}}+x_{\mathrm{aa}})y_{\mathrm{Aa}}] \\ x_{\mathrm{aa}}y_{\mathrm{aa}}+\frac{1}{2}(x_{\mathrm{aa}}y_{\mathrm{Aa}}+x_{\mathrm{Aa}}y_{\mathrm{aa}})+\frac{1}{4}x_{\mathrm{Aa}}y_{\mathrm{Aa}} \end{pmatrix}$$

$$=\begin{pmatrix} x_{\mathrm{AA}}+\frac{1}{2}x_{\mathrm{Aa}} & \frac{1}{2}x_{\mathrm{AA}}+\frac{1}{4}x_{\mathrm{Aa}} & 0 \\ \frac{1}{2}x_{\mathrm{Aa}}+x_{\mathrm{aa}} & \frac{1}{2}x_{\mathrm{AA}}+\frac{1}{2}x_{\mathrm{Aa}}+\frac{1}{2}x_{\mathrm{aa}} & x_{\mathrm{AA}}+\frac{1}{2}x_{\mathrm{Aa}} \\ 0 & \frac{1}{4}x_{\mathrm{Aa}}+\frac{1}{2}x_{\mathrm{aa}} & \frac{1}{2}x_{\mathrm{Aa}}+x_{\mathrm{aa}} \end{pmatrix}\begin{pmatrix} y_{\mathrm{AA}} \\ y_{\mathrm{Aa}} \\ y_{\mathrm{aa}} \end{pmatrix}=M_{\boldsymbol{x}}\boldsymbol{y}$$

$$M_{\boldsymbol{x}}$$

let's check our understanding so far

$$\boldsymbol{w},\ \boldsymbol{x},\ \boldsymbol{y},\ \boldsymbol{z} \in \mathcal{A}$$

$$M_{\boldsymbol{x}} : \mathcal{A} \to \mathcal{A}$$
$$\boldsymbol{w} \mapsto \boldsymbol{x}\boldsymbol{w}$$

$$(\boldsymbol{x}\boldsymbol{y})\boldsymbol{z} \stackrel{?}{=} \boldsymbol{x}(\boldsymbol{y}\boldsymbol{z})$$

No. Genetic algebras are not associative.

$$(M_{\boldsymbol{x}}M_{\boldsymbol{y}})M_{\boldsymbol{z}} \stackrel{?}{=} M_{\boldsymbol{x}}(M_{\boldsymbol{y}}M_{\boldsymbol{z}})$$

Yes. Transformation matrices are associative.

$$\boldsymbol{x}\boldsymbol{y} \stackrel{?}{=} \boldsymbol{y}\boldsymbol{x}$$

Yes. Genetic algebras are commutative.

$$M_{\boldsymbol{x}}M_{\boldsymbol{y}} \stackrel{?}{=} M_{\boldsymbol{y}}M_{\boldsymbol{x}}$$

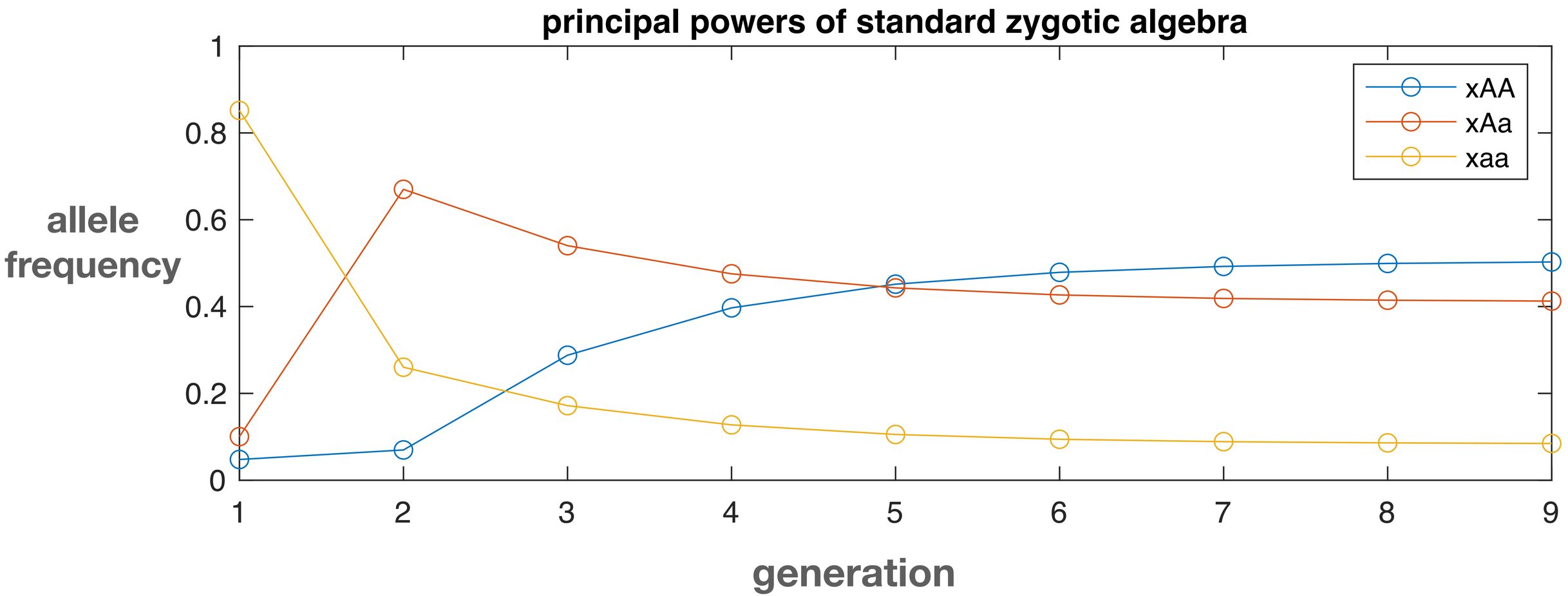
No. Transformation matrices do not commute.

$$\underbrace{M_{\boldsymbol{x}}M_{\boldsymbol{y}}\boldsymbol{z}}_{\boldsymbol{x}(\boldsymbol{y}\boldsymbol{z})} \stackrel{?}{=} \underbrace{M_{\boldsymbol{y}}M_{\boldsymbol{x}}\boldsymbol{z}}_{\boldsymbol{y}(\boldsymbol{x}\boldsymbol{z})}$$

Because the algebra is not associative.

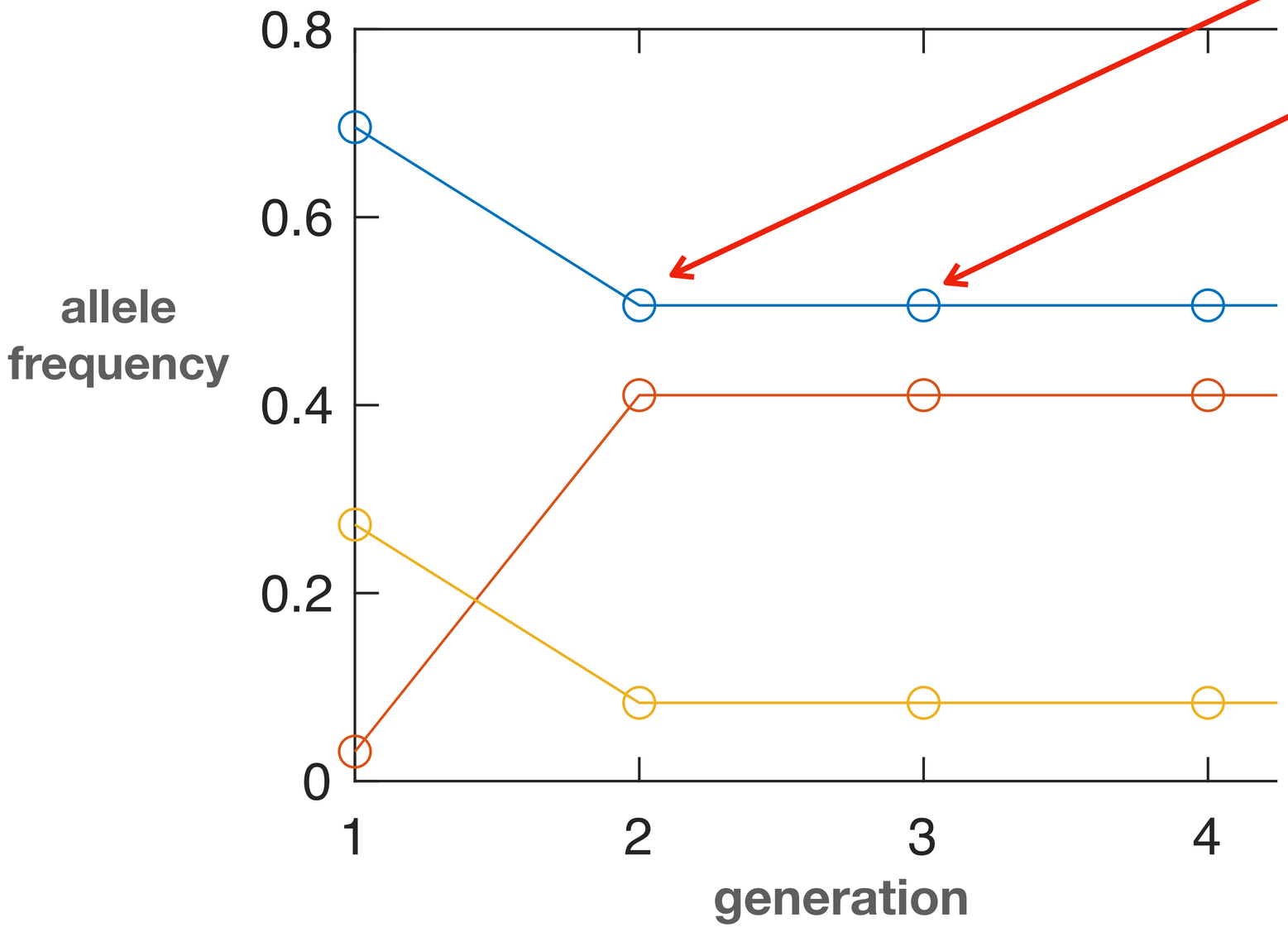
the transformation matrix provides numerical solution

$$\boldsymbol{a}_n = \boldsymbol{x}^n \boldsymbol{y} = M_{\boldsymbol{x}}^n \boldsymbol{y}$$



genetic_algebras.m

$$f(\boldsymbol{x}) = \boldsymbol{x}^3 - \boldsymbol{x}^2 = 0 \Longrightarrow \boldsymbol{x}^2 = (\boldsymbol{x}^2)^2$$



plenary powers

$$\boldsymbol{a}_n = \boldsymbol{x}^{2^n} = M_{\boldsymbol{a}_{n-1}} \boldsymbol{a}_{n-1}$$

genetic_algebras.m

the characteristic polynomial of the transformation matrix

$$c_M(\lambda) = \det(\lambda I - M) = \lambda^3 - \tfrac{3}{2}\lambda^2 + \tfrac{1}{2}\lambda = \lambda(\lambda - 1)(\lambda - \tfrac{1}{2})$$

rank equation

$$f(\boldsymbol{x}) \text{ divides } x\mathrm{c}_M(\boldsymbol{x})$$

characteristic polynomial

$$\boldsymbol{x}^2(\boldsymbol{x} - 1) \qquad \boldsymbol{x} \cdot \boldsymbol{x}(\boldsymbol{x} - 1)(\boldsymbol{x} - \tfrac{1}{2})$$

The eigenvalues of the transformation matrix *M* include the principal train roots of the rank equation *f(x)*, perhaps with repetition. The eigenvalues of *M* may also include 1/2 even if this is not a principal train root.

Genetic algebra illustrating train algebra identities

The transformation matrix M_x for the standard Mendelian zygotic algebra

In [1]:
```
var('p')
M = matrix(SR, 3, 3, [[p,p/2,0], [1-p,1/2,p], [0,(1-p)/2,1-p]])
M
```

Out[1]:
```
[           p        1/2*p            0]
[      -p + 1          1/2            p]
[           0 -1/2*p + 1/2       -p + 1]
```

In [2]:
```
M.eigenvalues()
```

Out[2]:
```
[1, 1/2, 0]
```

In [3]:
```
phi=M.minpoly(var='L')
phi
```

Out[3]:
```
L^3 - 3/2*L^2 + 1/2*L
```

In [4]:
```
phi(M).simplify_rational()
```

Out[4]:
```
[0 0 0]
[0 0 0]
[0 0 0]
```

The rank equation $r(x)$ is the minimal polynomial of the algebra

In [6]:
```
P.<x> = PolynomialRing(QQ,1)
r=x^3-x^2
r
```

Out[6]:
```
x^3 - x^2
```

Check rank equation by evaluating $f(M_x)x$ where $f(x) = r(x)/x$

In [7]:
```
var('a,b')
X = matrix(SR, 3, 1, [a, b, 1-a-b])
X # an arbitrary normalized element
```

Out[7]:
```
[         a]
[         b]
[-a - b + 1]
```

In [8]:
```
f=r/x
f
```

Out[8]:
```
x^2 - x
```

In [9]:
```
fM = f.subs(x=M).simplify_rational()
```

In [10]:
```
fMX = fM*X
fMX
```

Out[10]:
```
[                          -1/2*(a + b - 1)*p^2
[1/2*(2*p^2 - p)*(a + b - 1) - 1/2*(2*p^2 -
[   -1/2*(p^2 - p)*(a + b - 1) + 1/2*(p^2 -
```

In [11]:
```
fMX.subs(p=a+b/2).simplify_rational()
```

Out[11]:
```
[0]
[0]
[0]
```

gametic algebra for autotetraploids

	AA	Aa	aa
AA	AA	$\frac{1}{2}AA + \frac{1}{2}Aa$	$\frac{1}{6}AA + \frac{2}{3}Aa + \frac{1}{6}aa$
Aa	$\frac{1}{2}AA + \frac{1}{2}Aa$	$\frac{1}{6}AA + \frac{2}{3}Aa + \frac{1}{6}aa$	$\frac{1}{2}Aa + \frac{1}{2}aa$
aa	$\frac{1}{6}AA + \frac{2}{3}Aa + \frac{1}{6}aa$	$\frac{1}{2}Aa + \frac{1}{2}aa$	aa

6 ways to choose 2 from AAaa: AA ; Aa, Aa, Aa, Aa ; aa

change of basis helps with calculations

$$\mathsf{a} = \mathsf{AA},\ \mathsf{b} = \mathsf{AA} - \mathsf{Aa},\ \mathsf{c} = \mathsf{AA} - 2\mathsf{Aa} + \mathsf{aa}$$

	AA	Aa	aa
AA	AA	$\frac{1}{2}\mathsf{AA} + \frac{1}{2}\mathsf{Aa}$	$\frac{1}{6}\mathsf{AA} + \frac{2}{3}\mathsf{Aa} + \frac{1}{6}\mathsf{aa}$
Aa	$\star$	$\frac{1}{6}\mathsf{AA} + \frac{2}{3}\mathsf{Aa} + \frac{1}{6}\mathsf{aa}$	$\frac{1}{2}\mathsf{Aa} + \frac{1}{2}\mathsf{aa}$
aa	$\star$	$\star$	aa

	a	b	c
a	a	$\frac{1}{2}\mathsf{b}$	$\frac{1}{6}\mathsf{c}$
b	$\frac{1}{2}\mathsf{b}$	$\frac{1}{6}\mathsf{c}$	0
c	$\frac{1}{6}\mathsf{c}$	0	0

$$\mathsf{ab} = (\mathsf{AA})(\mathsf{AA} - \mathsf{Aa}) = \mathsf{AA} - \left(\tfrac{1}{2}\mathsf{AA} + \tfrac{1}{2}\mathsf{Aa}\right) = \tfrac{1}{2}\mathsf{AA} - \tfrac{1}{2}\mathsf{Aa} = \tfrac{1}{2}\mathsf{a} - \tfrac{1}{2}\left(\mathsf{a} - \mathsf{b}\right) = \tfrac{1}{2}\mathsf{b}$$

change of basis helps with calculations

	a	$-x_b\mathsf{b}$	$x_c\mathsf{c}$
a	a	$-\frac{1}{2}x_b\mathsf{b}$	$\frac{1}{6}x_c\mathsf{c}$
$-y_b\mathsf{b}$	$-\frac{1}{2}y_b\mathsf{b}$	$\frac{1}{6}y_bx_b\mathsf{c}$	0
$y_c\mathsf{c}$	$\frac{1}{6}y_c\mathsf{c}$	0	0

$$\begin{pmatrix} 1 \\ -\frac{1}{2}(y_b + x_b) \\ \frac{1}{6}(x_c + y_bx_b + y_c) \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ -\frac{1}{2}x_b & \frac{1}{2} & 0 \\ \frac{1}{6}x_c & -\frac{1}{6}x_b & \frac{1}{6} \end{pmatrix}}_{M_{\boldsymbol{x}}} \begin{pmatrix} 1 \\ -y_b \\ y_c \end{pmatrix}$$

$$c_M(\lambda) = (\lambda - 1)(\lambda - \tfrac{1}{2})(\lambda - \tfrac{1}{6})$$

Because $f(x)$ must divide $xc_M(x)$, I guess that the rank equation is

$$f(x) = x(x-1)(x-\tfrac{1}{6}) = 0$$

with principle train roots $\lambda_0 = 1$ and $\lambda_1 = \frac{1}{6}$.

To check this, note $\boldsymbol{x} = \mathsf{a} - b\mathsf{b} + c\mathsf{c}$ and calculate

$$\boldsymbol{x}^2 = \mathsf{a} - b\mathsf{b} + \tfrac{1}{6}(b^2 + 2c)\mathsf{c}\,.$$

Subtracting $\quad \boldsymbol{x}(\boldsymbol{x}-1) = (\frac{1}{6}b^2 - \frac{2}{3}c)\mathsf{c}$

Multiplying by $\boldsymbol{x}$

$$\boldsymbol{x}(\boldsymbol{x}-1)\boldsymbol{x} = (\tfrac{1}{6}b^2 - \tfrac{2}{3}c)\mathsf{c}\cdot(\mathsf{a} - b\mathsf{b} + c\mathsf{c}) = \tfrac{1}{6}(\tfrac{1}{6}b^2 - \tfrac{2}{3}c)\mathsf{c} = \boldsymbol{x}(\boldsymbol{x}-1)\tfrac{1}{6}.$$

Subtracting $\quad x(x-1)(x-\frac{1}{6}) = 0$ ✓

principle powers

	a	$-x_b$b	x_cc
a	a	$-\frac{1}{2}x_b$b	$\frac{1}{6}x_c$c
$-y_b$b	$-\frac{1}{2}y_b$b	$\frac{1}{6}y_bx_b$c	0
y_cc	$\frac{1}{6}y_c$c	0	0

$$\begin{pmatrix} 1 \\ -\frac{1}{2}(y_b + x_b) \\ \frac{1}{6}(x_c + y_b x_b + y_c) \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ -\frac{1}{2}x_b & \frac{1}{2} & 0 \\ \frac{1}{6}x_c & -\frac{1}{6}x_b & \frac{1}{6} \end{pmatrix}}_{M_{\boldsymbol{x}}} \begin{pmatrix} 1 \\ -y_b \\ y_c \end{pmatrix}$$

$$c_M(\lambda) = (\lambda - 1)(\lambda - \tfrac{1}{2})(\lambda - \tfrac{1}{6})$$

$$f(x) = x(x-1)(x-\tfrac{1}{6}) = 0$$

$$\boldsymbol{x}^3 = \tfrac{7}{6}\boldsymbol{x}^2 - \tfrac{1}{6}\boldsymbol{x}$$

plenary powers

$\boldsymbol{x}^2$	a	$-b$b	cc
a	a	$-\frac{1}{2}b$b	$\frac{1}{6}c$c
$-b$b	$-\frac{1}{2}b$b	$\frac{1}{6}b^2$c	0
cc	$\frac{1}{6}c$c	0	0

$$\underbrace{\begin{pmatrix} 1 \\ -b \\ \frac{1}{6}(b^2 + 2c) \end{pmatrix}}_{\boldsymbol{x}^2} = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -\frac{1}{6}b & \frac{1}{3} \end{pmatrix}}_{M_{\boldsymbol{x}}} \underbrace{\begin{pmatrix} 1 \\ -b \\ c \end{pmatrix}}_{\boldsymbol{x}}$$

$$c_M(\lambda) = (\lambda - 1)^2(\lambda - \tfrac{1}{3})$$

$$\begin{aligned} f(x) &= x[x-1][x-\tfrac{1}{3}] \\ &= (x^2 - x)[x - \tfrac{1}{3}] \\ &= ((x^2)^2 - x^2) - \tfrac{1}{3}(x^2 - x) \end{aligned}$$

$$(\boldsymbol{x}^2)^2 = \tfrac{4}{3}\boldsymbol{x}^2 - \tfrac{4}{3}\boldsymbol{x}$$

gametic autotetraploid algebra

(1) using abc basis found transformation matrix M

(2) iterative solution using M $\boldsymbol{a}_n = \boldsymbol{x}^n \boldsymbol{y} = M_{\boldsymbol{x}}^n \boldsymbol{y}$

(3) rank equation

$$f(x) = x(x-1)(x-\tfrac{1}{6}) = x^3 - \tfrac{7}{6}x^2 + \tfrac{1}{6}x$$

$$\text{principle train roots } \lambda_0 = 1 \text{ and } \lambda_1 = \tfrac{1}{6}$$

(4) recurrence equation

$$\boldsymbol{a}_0 = \boldsymbol{y}\,, \quad \boldsymbol{a}_1 = \boldsymbol{x} \cdot \boldsymbol{y}\,, \quad \boldsymbol{a}_n = \tfrac{7}{6}\boldsymbol{a}_{n-1} - \tfrac{1}{6}\boldsymbol{a}_{n-2} \quad \text{for} \quad n \geq 2$$

(5) can be generalized to *n*-ploid gametic algebra

We shall first consider the gametic n-ploidy algebra. The types will be denoted by $D_n, D_{n-1}, \ldots, D_0$ where D_i stands for the type which has i dominant genes and $n-i$ recessive genes. The multiplication table is:

$$D_a D_b = \binom{2n}{n}^{-1} \sum_{i=0}^{n} \binom{a+b}{i}\binom{2n-a-b}{n-i} D_i$$

$$c_a = \sum_{i=0}^{a} (-1)^i \binom{a}{i} D_i \qquad c_a c_b = \binom{2n}{n}^{-1} \binom{2n-m}{n-m} c_m.$$

train roots $\binom{2n}{i}^{-1}\binom{n}{i}$ *for* $0 \leqq i \leqq n$ $\qquad \lambda_0 = 1,\ \lambda_1 = \frac{1}{2}$

all other λ's less than $\frac{1}{2}$

```
                                1
                            1       1
                        1       2       1
                    1       3       3       1
                1       4       6       4       1
            1       5      10      10       5       1
        1       6      15      20      15       6       1
    1       7      21      35      35      21       7       1
1       8      28      56      70      56      28       8       1
    9      36      84     126     126      84      36       9       1
   10      45     120     210     252     210     120      45      10       1
   11      55     165     330     462     462     330     165      55      11       1
   12      66     220     495     792     924     792     495     220      66      12       1
```

n=2: 1, 1/2, 1/6

n=4: 1, 1/2, 3/14, 1/14, 1/70

n=6: 1/1, 6/12, 15/66, 20/220, 15/495, 6/792, 1/924

Gonshor

more realism

(1) mutation and selection

(2) linkage disequilibrium & recombination

(3) sex linked inheritance XX XY

(4) mating types and sexual selection

zygotic X chromosome-linked inheritance

		XX			XY	
		AA	Aa	aa	AY	aY
XX	AA	0	0	0	$\frac{1}{2}AA + \frac{1}{2}AY$	$\frac{1}{2}Aa + \frac{1}{2}AY$
	Aa		0	0	$\frac{1}{4}AA + \frac{1}{4}Aa + \frac{1}{4}AY + \frac{1}{4}aY$	$\frac{1}{4}Aa + \frac{1}{4}aa + \frac{1}{4}AY + \frac{1}{4}aY$
	aa			0	$\frac{1}{2}Aa + \frac{1}{2}aY$	$\frac{1}{2}aa + \frac{1}{2}aY$
XY	AY				0	0
	aY					0

communative duplication of genetic algebras

realized using tensor products

$$\mathcal{Z} = \mathcal{G} \otimes \mathcal{G}$$

unordered pairs of gametes (parental types)

$$\mathcal{K} = \mathcal{Z} \otimes \mathcal{Z}$$

unordered pairs of zygotes (mating types)

$\mathcal{G}$

	A	a
A	A	$\frac{1}{2}(A+a)$
a	$\frac{1}{2}(a+A)$	a

Gametic algebra

	AA	Aa	aa
AA	AA	$\frac{1}{2}(AA+Aa)$	Aa
Aa	$\frac{1}{2}(AA+Aa)$	$\frac{1}{4}AA+\frac{1}{2}Aa+\frac{1}{4}aa$	$\frac{1}{2}(Aa+aa)$
aa	Aa	$\frac{1}{2}(Aa+aa)$	aa

Zygotic algebra

	AA|AA	AA|Aa	AA|aa	Aa|Aa	Aa|aa	aa|aa
AA|AA						
AA|Aa						
AA|aa						
Aa|Aa						
Aa|aa						
aa|aa						

Copular algebra

zygotic from gametic autotetraploid algebra

The zygotic autotetraploid algebra has $\binom{3}{2}$ basis elements:

$$\mathsf{AA} \otimes \mathsf{AA}\,, \quad \mathsf{AA} \otimes \mathsf{Aa}\,, \quad \mathsf{AA} \otimes \mathsf{aa}\,, \quad \mathsf{Aa} \otimes \mathsf{Aa}\,, \quad \mathsf{Aa} \otimes \mathsf{aa}\,, \quad \mathsf{aa} \otimes \mathsf{aa}\,.$$

where $x \otimes y = y \otimes x$. The products of tensors are defined using $(w \otimes x)(y \otimes z) = (wy \otimes xz)$, for example:

$$\begin{aligned}
(\mathsf{AA} \otimes \mathsf{Aa})(\mathsf{aa} \otimes \mathsf{AA}) &= (\mathsf{AA} \cdot \mathsf{aa}) \otimes (\mathsf{Aa} \cdot \mathsf{AA}) \\
&= (\tfrac{1}{6}\mathsf{AA} + \tfrac{2}{3}\mathsf{Aa} + \tfrac{1}{6}\mathsf{aa}) \otimes (\tfrac{1}{2}\mathsf{AA} + \tfrac{1}{2}\mathsf{Aa}) \\
&= \tfrac{1}{12}\mathsf{AA} \otimes \mathsf{AA} + \tfrac{1}{3}\mathsf{Aa} \otimes \mathsf{AA} + \tfrac{1}{12}\mathsf{aa} \otimes \mathsf{AA} + \tfrac{1}{12}\mathsf{AA} \otimes \mathsf{Aa} + \tfrac{1}{3}\mathsf{Aa} \otimes \mathsf{Aa} + \tfrac{1}{12}\mathsf{aa} \otimes \mathsf{Aa} \\
&= \tfrac{1}{12}\mathsf{AA} \otimes \mathsf{AA} + \tfrac{5}{12}\mathsf{AA} \otimes \mathsf{Aa} + \tfrac{1}{12}\mathsf{AA} \otimes \mathsf{aa} + + \tfrac{1}{3}\mathsf{Aa} \otimes \mathsf{Aa} + \tfrac{1}{12}\mathsf{Aa} \otimes \mathsf{aa}\,.
\end{aligned}$$

multiplication tables in the abc basis

gametic

	a	b	c
a	a	$\frac{1}{2}$b	$\frac{1}{6}$c
b	$\frac{1}{2}$b	$\frac{1}{6}$c	0
c	$\frac{1}{6}$c	0	0

zygotic

	a ⊗ a	a ⊗ b	a ⊗ c	b ⊗ b	b ⊗ c	c ⊗ c
a ⊗ a	a ⊗ a	$\frac{1}{2}$a ⊗ b	$\frac{1}{6}$a ⊗ c	$\frac{1}{4}$b ⊗ b	$\frac{1}{12}$b ⊗ c	$\frac{1}{36}$c ⊗ c
a ⊗ b	$\star$	$\frac{1}{6}$a ⊗ c	0	0	0	0
a ⊗ c	$\star$	$\star$	0	0	0	0
b ⊗ b	$\star$	$\star$	$\star$	$\frac{1}{36}$c ⊗ c	0	0
b ⊗ c	$\star$	$\star$	$\star$	$\star$	0	0
c ⊗ c	$\star$	$\star$	$\star$	$\star$	$\star$	0

mathematical and computational aspects

(1) transformation matrix from structure constants

(2) the transformation algebra

(3) canonical basis of a genetic algebra

(4) implementing a change of basis

(5) Haldane linearization

(6) classification of genetic algebras

transformation matrix from structure constants

$$\boldsymbol{x}\cdot\boldsymbol{y}=\sum_{ij}x_iy_j\,\mathsf{a}_i\mathsf{a}_j=\sum_{ij}x_iy_j\sum_k\gamma_{ij}^k\mathsf{a}_k\,.$$

The a_ℓ component of $\boldsymbol{x}\cdot\boldsymbol{y}$ is

$$[\boldsymbol{x}\cdot\boldsymbol{y}]_\ell=\sum_{ij}x_iy_j\gamma_{ij}^\ell\,.$$

As a linear operator we have $\boldsymbol{x}\cdot\boldsymbol{y}=M_{\boldsymbol{x}}\boldsymbol{y}$. For the a_ℓ component,

$$[\boldsymbol{x}\cdot\boldsymbol{y}]_\ell=\sum_j M_{\ell j}y_j\,.$$

Equate these two expressions for $[\boldsymbol{x}\cdot\boldsymbol{y}]_\ell$ gives

$$\sum_j M_{\ell j}y_j=\sum_{ij}x_iy_j\gamma_{ij}^\ell\,.$$

The equality is term by term so for a given y_j,

$$M_{\ell j}=\sum_i x_i\gamma_{ij}^\ell\,.$$

canonical basis of a genetic algebra

$$\mathsf{a}_i\mathsf{a}_j = \sum_k \gamma_{ij}^k \mathsf{a}_k$$

$$\gamma_{00}^0 = 1 \qquad \gamma_{ij}^k = \gamma_{ji}^k$$

$$\gamma_{0j}^k = 0 \quad \text{for} \quad k < j$$

$$i, j > 0 \quad \gamma_{ij}^k = 0 \quad \text{for} \quad k \leq \max(i, j)$$

	a_0	a_1	a_2
a_0	$(1)\mathsf{a}_0 + \gamma_{00}^1\mathsf{a}_1 + \gamma_{00}^2\mathsf{a}_2 + \cdots$	$\gamma_{01}^1\mathsf{a}_1 + \gamma_{01}^2\mathsf{a}_2 + \gamma_{01}^3\mathsf{a}_3 + \cdots$	$\gamma_{02}^2\mathsf{a}_2 + \gamma_{02}^3\mathsf{a}_3 + \gamma_{02}^4\mathsf{a}_4 + \cdots$
a_1	$\star$	$\gamma_{11}^2\mathsf{a}_2 + \gamma_{11}^3\mathsf{a}_3 + \cdots$	$\gamma_{12}^2\mathsf{a}_2 + \gamma_{12}^3\mathsf{a}_3 + \gamma_{12}^4\mathsf{a}_4 + \cdots$
a_2	$\star$	$\star$	$\gamma_{22}^3\mathsf{a}_3 + \gamma_{22}^4\mathsf{a}_4 + \cdots$

allowed	a_0	a_1	a_2
a_0	$\mathsf{a}_0\mathsf{a}_1\mathsf{a}_2$	$\mathsf{a}_1\mathsf{a}_2$	a_2
a_1	$\mathsf{a}_1\mathsf{a}_2$	a_2	0
a_2	a_2	0	0

implementing a change of basis

$$b_i = \sum_k t_{ki} a_k$$

$$a_i a_j = \sum_k \gamma_{ij}^k a_k \qquad\qquad b_i b_j = \sum_k g_{ij}^k b_k$$

$$g_{ij}^n = \sum_{k\ell m} t_{ki} t_{\ell j} \gamma_{k\ell}^m t_{nm}^{-1}$$

In general, the plenary powers map $E: \mathbf{p} \to \mathbf{p}^2$ is NOT linear

$$\mathbf{p} = \mathbf{w}_0 + g^1\mathbf{w}_1 + g^2\mathbf{w}_2$$

	$\mathbf{w}_0$	$\mathbf{w}_1$	$\mathbf{w}_2$
$\mathbf{w}_0$	$\mathbf{w}_0 + t^1_{00}\mathbf{w}_1 + t^2_{00}\mathbf{w}_2$	$t^1_{01}\mathbf{w}_1 + t^2_{01}\mathbf{w}_2$	$t^2_{02}\mathbf{w}_2$
$\mathbf{w}_1$		$t^2_{11}\mathbf{w}_2$	$\mathbf{0}$
$\mathbf{w}_2$			$\mathbf{0}$

$$g^1 \longrightarrow t^1_{00} + 2t^1_{01}g^1 ,$$

$$g^2 \longrightarrow t^2_{00} + 2t^2_{01}g^1 + 2t^2_{02}g^2 + t^2_{11}(g^1)^2$$

$$(g^1)^2 \longrightarrow (t^1_{00} + 2t^1_{01}g^1)^2$$
$$= (t^1_{00})^2 + 4t^1_{00}t^1_{01}g^1 + 4(t^1_{01})^2(g^1)^2$$

$$\begin{array}{ccc} (g^1, g^2) & & (1, g^1, (g^1)^2, g^2, \ldots) \\ \mathbf{p} & \longrightarrow & \mathbf{W} \\ \text{quadratic } E \;\downarrow & & \downarrow L \text{ linear} \\ (g^1, g^2)\ \mathbf{p}^2 & \longrightarrow & \mathbf{W}\ (1, g^1, (g^1)^2, g^2, \ldots) \end{array}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ t^1_{00} & 2t^1_{01} & 0 & 0 \\ (t^1_{00})^2 & 4t^1_{00}t^1_{01} & 4(t^1_{01})^2 & 0 \\ t^2_{00} & 2t^2_{01} & t^2_{11} & 2t^2_{02} \end{bmatrix}$$

The quadratic operator can be represented by a linear transformation on a higher dimensional space

Algebras with genetic realization

$\Uparrow$

Baric algebras

$\Uparrow$

Train algebras

$\Uparrow$

Genetic algebras

(Schafer = Gonshor)

$\Uparrow$

Special train algebras

structure constant constraint (table preserves normalization)

$$\boldsymbol{a}_i\boldsymbol{a}_j = \sum_{k=1}^{n} \gamma_{ij}^k \boldsymbol{a}_k \qquad 0 \le \gamma_{ij}^k \le 1 \qquad \sum_{k=1}^{n} \gamma_{ij}^k = 1$$

1 dimensional representation (weight function)

$$\omega \colon A \to \mathbb{R} \text{ by } \omega(a_i) = 1 \qquad \omega(xy) = \omega(x)\omega(y)$$

$$\boldsymbol{x} = \sum_i x_i \boldsymbol{a}_i \qquad \omega(\boldsymbol{x}) = \sum_i x_i \omega(\mathsf{a}_i) = 1$$

rank function constraint

$$f(x) = x^r + \theta_1 x^{r-1} + \theta_2 x^{r-2} + \cdots + \theta_{r-2}x^2 + \theta_{r-1}x$$

θ_i depend only on the $\omega(\mathsf{a}_i)$ not on the x_i

tranformation algebra constraint

for any transformation $T = \alpha I + f(M_{\boldsymbol{x}}, M_{\boldsymbol{y}}, \ldots)$

coefficients of $c_T(\lambda) = \det(\lambda I - T)$

depend only on $\omega(\boldsymbol{x})$, $\omega(\boldsymbol{y})$, ...

principal power subalgebra constraint

$N = Ker\,\omega$ is nilpotent

$$A \supset N = N^1 \supset N^2 \supset N^3 \supset \cdots N^r \supset N^{r+1} = < 0 >$$

references and further reading

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ALGEBRAIC STRUCTURE OF GENETIC INHERITANCE

MARY LYNN REED

ABSTRACT. In this paper we will explore the nonassociative algebraic structure that naturally occurs as genetic information gets passed down through the generations. While modern understanding of genetic inheritance initiated with the theories of Charles Darwin, it was the Augustinian monk Gregor Mendel who began to uncover the mathematical nature of the subject. In fact, the symbolism Mendel used to describe his first results (e.g., see his 1866 paper *Experiments in Plant-Hybridization* [30]) is quite algebraically suggestive. Seventy four years later, I.M.H. Etherington introduced the formal language of abstract algebra to the study of genetics in his series of seminal papers [9], [10], [11]. In this paper we will discuss the concepts of genetics that suggest the underlying algebraic structure of inheritance, and we will give a brief overview of the algebras which arise in genetics and some of their basic properties and relationships. With the popularity of biologically motivated mathematics continuing to rise, we offer this survey article as another example of the breadth of mathematics that has biological significance. The most comprehensive reference for the mathematical research done in this area (through 1980) is Wörz-Busekros [36].

1. GENETIC MOTIVATION

Before we discuss the mathematics of genetics, we need to acquaint ourselves with the necessary language from biology. A vague, but nevertheless informative, definition of a **gene** is simply a unit of hereditary information. The genetic code of an organism is carried on **chromosomes**. Each gene on a chromosome has different forms that it can take. These forms are called **alleles**. E.g., the gene which determines blood type in humans has three different alleles, A, B, and O. Since humans are **diploid** organisms (meaning we carry a double set of chromosomes – one from each parent), blood types are determined by two alleles. **Haploid** cells (or organisms) carry a single set of chromosomes. When diploid organisms reproduce, a process called **meiosis** produces **gametes** (sex cells) which carry a single set of chromosomes. When these gamete cells fuse (e.g., when sperm fertilizes egg), the result is a **zygote**, which is again a diploid cell, meaning it carries its hereditary information in a double set of chromosomes. When gametes fuse (or reproduce) to form zygotes a natural "multiplication" operation occurs.

1.1. **Simple Mendelian Inheritance.** As a natural first example, we consider simple Mendelian inheritance for a single gene with two alleles A and a. In this case, two gametes fusing (or reproducing) to form a zygote gives the multiplication table shown in Table 1, which in freshman biology class might be called a Punnett square.

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XXIII.—**Genetic Algebras.** By **I. M. H. Etherington,** B.A.(Oxon), Ph.D.(Edin.), Mathematical Institute, University of Edinburgh.

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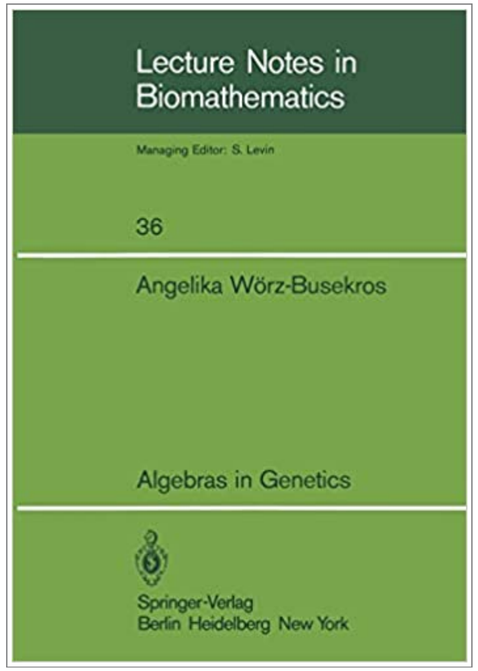
§ 1. INTRODUCTION.

TWO classes of linear algebras, generally non-associative, are defined in § 3 (*baric algebras*) and § 4 (*train algebras*), and the process of *duplication* of a linear algebra in § 5. These concepts, which will be discussed more fully elsewhere, arise naturally in the symbolism of genetics, as shown in §§ 6–15. Many of their properties express facts well known in genetics; and the processes of calculation which are fundamental in many problems of population genetics can be expressed as manipulations in the genetic algebras. In cases where inheritance is of a simple type (*e.g.* §§ 10–13, 15) this constitutes a new point of view, but perhaps amounts to little more than a change of notation as compared with existing methods. § 14, however, indicates the possibility of generalisations which would seem to be impossible by ordinary methods.

The occurrence of the genetic algebras may be described in general terms as follows. The mechanism of chromosome inheritance, in so far as it determines the probability distributions of genetic types in families and filial generations, and expresses itself through their frequency distributions, may be represented conveniently by algebraic symbols. Such a symbolism is described, for instance, by Jennings (1935, chap. ix);



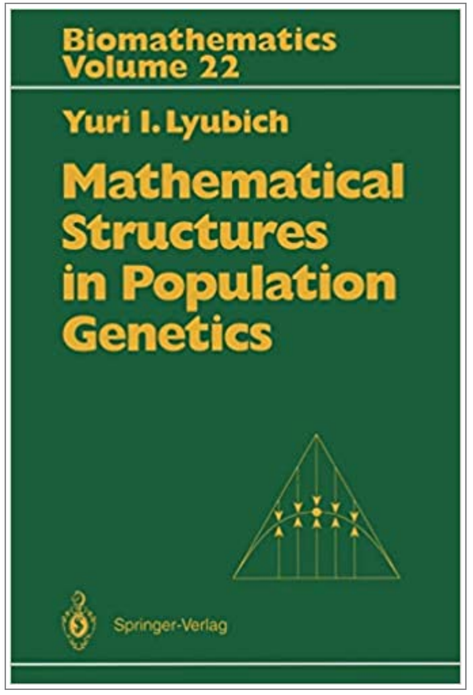
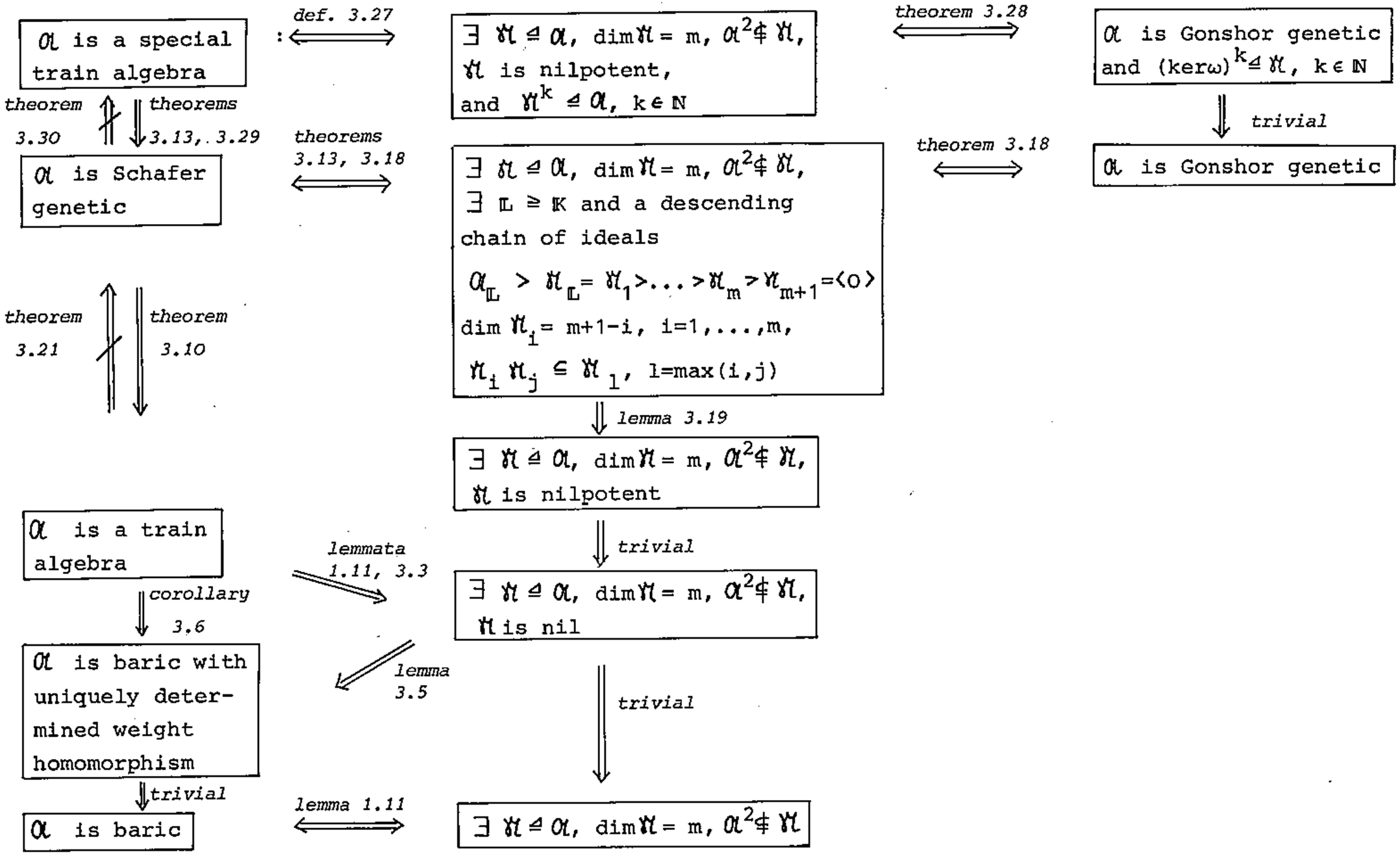


Table 3.31. Let $\mathcal{A}$ be an (m+1)-dimensional commutative algebra over $\mathbb{K}$



dimension 3 genetic algebras

canonical basis

	c_o	c_1	c_2
c_o	c_o	$\lambda_1 c_1 + \alpha c_2$	$\lambda_2 c_2$
c_1		βc_2	0
c_2			0

$1, \lambda_1, \lambda_2$ are the train roots

λ_1, λ_2	α, β	idempotents	rank equation for x $\omega(x)=1$
$\lambda_1, \lambda_2 \neq \frac{1}{2}$ $\lambda_1 \neq \lambda_2$	arbitrary	c_o	$x(x-1)(x-\lambda_1)(x-\lambda_2)=0$
$\lambda_1=\lambda_2 \neq \frac{1}{2}$	$(\alpha,\beta)\neq(0,0)$	c_o	$x(x-1)(x-\lambda_1)^2=0$
	$(\alpha,\beta)=(0,0)$	c_o	$x(x-1)(x-\lambda_1)=0$
$\lambda_1 = \frac{1}{2}$ $\lambda_2 \neq \frac{1}{2}$	$\alpha=0, \beta=0$	$\mathcal{E}_1=\{c_o+\eta_1 c_1 \mid \eta_1 \in \mathbb{L}\}$	$x(x-1)(x-\lambda_2) = 0$
	$\alpha\neq 0, \beta=0$	$\mathcal{E}_1^*=\{c_o+\eta_1 c_1 + \eta_2 c_2 \mid \eta_1 \in \mathbb{L}, \eta_2 = \frac{2\alpha}{1-2\lambda_2}\eta_1\}$	
	α arbitrary $\beta \neq 0$	$\mathcal{E}_\gamma=\{c_o+\eta_1 c_1+\eta_2 c_2 \mid \eta_1 \in \mathbb{L}, \eta_2 = \frac{2\alpha\eta_1+\beta\eta_1^2}{1-2\lambda_2}\}$	
$\lambda_1 = \frac{1}{2}$ $\lambda_2 = \frac{1}{2}$	$\alpha=0, \beta=0$	$\mathcal{E}=\{c_o+\eta_1 c_1+\eta_2 c_2 \mid \eta_1, \eta_2 \in \mathbb{L}\}$	$x(x-1) = 0$
	$\alpha\neq 0, \beta=0$ $\alpha=0, \beta\neq 0$	$\mathcal{E}_2=\{c_o+\eta_2 c_2 \mid \eta_2 \in \mathbb{L}\}$	$x(x-1)(x-\frac{1}{2}) = 0$
	$\alpha\neq 0, \beta\neq 0$	$\mathcal{E}_2=\{c_o+\eta_2 c_2 \mid \eta_2 \in \mathbb{L}$ $\mathcal{E}_2^*=\{c_o-\frac{2\alpha}{\beta}c_1+\eta_2 c_2 \mid \eta_2 \in \mathbb{L}\}$	
$\lambda_1 \neq \frac{1}{2}$ $\lambda_2 = \frac{1}{2}$	α arbitrary $\beta \neq 0$	$\mathcal{E}_2=\{c_o+\eta_2 c_2 \mid \eta_2 \in \mathbb{L}\}$	$x(x-1)(x-\lambda_1)(x-\frac{1}{2})=0$
	α arbitrary $\beta = 0$		$x(x-1)(x-\lambda_1)=0$

constants of the general solution

$$\boldsymbol{a}_n = c_0\boldsymbol{v}_0(0)^n + c_1\boldsymbol{v}_1(1)^n + c_{1/6}\boldsymbol{v}_{1/6}(1/6)^n$$

$$\boldsymbol{a}_0 = \boldsymbol{y} \qquad \boldsymbol{a}_0 = c_0\boldsymbol{v}_0 + c_1\boldsymbol{v}_1 + c_{1/6}\boldsymbol{v}_{1/6}$$

$$\boldsymbol{a}_1 = \boldsymbol{xy} \qquad \boldsymbol{a}_1 = c_1\boldsymbol{v}_1 + c_{1/6}\boldsymbol{v}_{1/6}(1/6)$$